

Adaptive Control of Thought

The Basic Idea

The Adaptive Control of Thought (ACT) is a theory about cognition and memory. Over the years, it has evolved and its most recent version is iterated as ACT-R, with the added letter standing in for 'Rational'. According to ACT theory, we can create a model of the human brain through which we can analyze and predict human behavior, most recently based on the idea that people act rationally. ACT-R is therefore a cognitive architecture that maps out the functions of our higher cognitive processes to demonstrate how people process information and then act accordingly.¹

ACT posits that multiple higher cognitive processes have the same underlying system; all of our thoughts, no matter what kind of thought it is, arises as a result of the same brain function. One of its most central tenets is that knowledge be divided into two kinds: declarative knowledge and procedural knowledge.²

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We know that human agents tend to be generally well-adapted to their environment, and hence a careful analysis of the cognitive task encountered by the mind, coupled with an assumption of the optimality of human behavior, results in a putatively powerful methodology of prediction and explanation.

– Samuli Reijula, professor in philosophy at the University of Helsinki, in his paper *The Upside of Irrationality: The Unexpected Benefits of Defying Logic at Work and at Home* “What, when and how do rational analysis models explain?”¹

Key Terms

Declarative Knowledge: Declarative knowledge (also known as descriptive knowledge) is knowledge that we are conscious of and can verbalize. It is information we directly encode from the environment and doesn't require much synthesization.³ It emphasizes *what* one needs to do to solve an issue rather than *how* to solve it. Examples of declarative knowledge include facts, world history, or rules for solving mathematical equations.⁴ In essence, it is content that can be recited or memorized.

Procedural Knowledge: Procedural knowledge (also known as imperative knowledge) is knowledge you use while performing a task, but may not be able to verbalize. It is information encoded from synthesizing and observing transformations of the environment (behaviors).³ Procedural knowledge is about *how* we do something. Examples of procedural knowledge include behaviors we do [habitually](#), such as riding a bike or driving a car.

Human Information Processing: Insights into how people receive, store, integrate, retrieve and apply information; in other words, how people learn. ACT is a theory of human information processing.⁵

Serial Processing: Serial processing suggests that only one piece of information can be processed at a time and that we encode information one object after the other.⁶

Parallel Processing: Parallel processing suggests that multiple pieces of information can be processed at a time and that we encode objects simultaneously.⁶

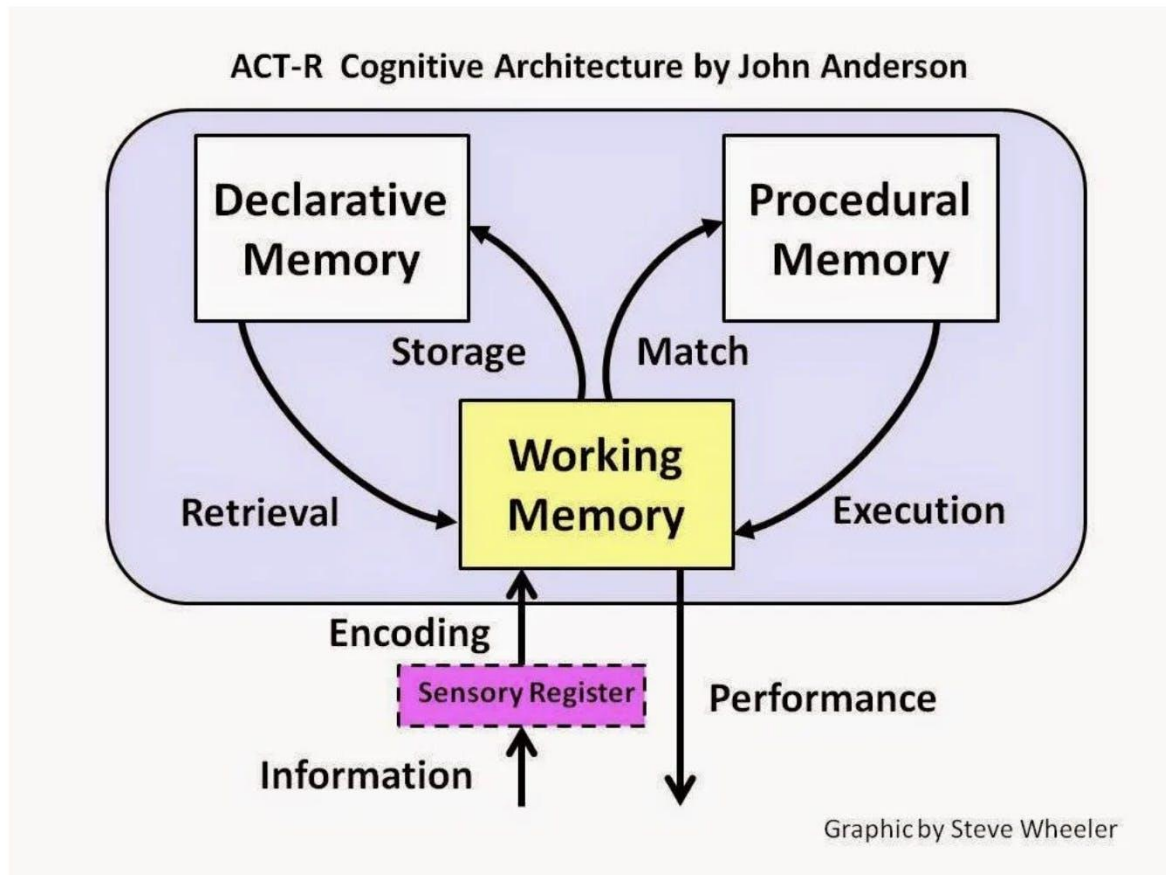
Rational Analysis: Rational analysis is the assumption that people react rationally to their environment. Traditional economics uses rational analysis to predict consumer behavior.⁷

History

Canadian psychologist John Anderson has an extensive background researching and developing his Adaptive Control of Thought model. His underlying assumption is that knowledge can be reduced into a theory or model, and therefore that a system can be created to perform human cognitive tasks.⁸ This assumption rests on the belief, articulated by Anderson in his 1990 book, that “*we can understand a lot about human cognition without considering in detail what is inside the human head. Rather, we can look in detail at what is outside the human head and try to determine what would be optimal behavior given the structure of the environment and the goals of the human*” (3)¹. Such systems operate based on rules like ‘If x, then y.’

With the goal in mind of creating a model to depict human knowledge, Anderson developed the Human Associative Memory (HAM) model alongside cognitive psychologist Gordon Bower in 1973. This model computed the mathematical theories of human cognition prevalent in the 1950s and 1960s. However, the model only accounted for human memory and did not accomplish Anderson’s mission of showing that all higher cognitive processes (memory, language, problem solving, imagery, deduction and induction) have the same underlying system. Replacing the HAM model, Anderson developed the ACT model in 1976, which was able to account for these higher cognitive processes.⁹

In 1990, Anderson developed another version of ACT which he named ACT*. His hypothesis was that ACT*, alike ACT, showed that the mind is unitary (all of thoughts and mental faculties can be explained by the same underlying system) and that our experiences are stored in different facilities (linguistic, geometric, etc.) depending on their subject matter.¹⁰ Information comes through a ‘buffer’, known as working memory, which determines whether the information should be stored as declarative and later retrieved, or as procedural knowledge and executed in the moment to match current activity. The model looks as follows:²



The original ACT model only accounted for one kind of ‘cognitive unit’ that the mind would process, also known as ‘chunks’. These units were words and statements like ‘hate’ or ‘my mom is nice’. and Anderson speculated that working memory is able to process around 5 chunks at once before becoming overwhelmed.

However, in the ACT* model, information could come in the form of spatial images and temporal strings as well as abstract propositions.¹⁰ What counted as a cognitive unit was expanded. Temporal strings encode the order of a set of items; spatial images encode the spatial configuration of a cognitive unit; abstract propositions encode meaning. Another key element of ACT* was its suggestion that all information begins as declarative, and combines with method learning to produce procedural knowledge.¹¹

Moreover, ACT suggests that information was serially processed, whereas ACT* suggests information could be processed simultaneously (parallel processing). The last notable difference is that the ACT* theory added a ‘sub-symbolic’ component: a feature that could determine what meaning is activated when we run into a cognitive unit with many possibilities. For example, if we encounter the sentence “*The robber took money from the bank,*” the word ‘bank’ has two meanings: a financial institution, or the land sloping down next to a body of water. The sub-symbolic component of ACT* activates the financial institution meaning because it knows that meaning is processed in relation to money or robbery.¹⁰

A few years later, Anderson, alongside his colleagues at Carnegie Mellon University, developed the most recent model of ACT by combining the original with rational analysis. This model is known as ACT-R and predicts behavior based on the idea that humans act in ‘optimal’ (see: rational) ways. The ACT-R is essentially a production system, like a machine, that operates according to ACT*. The theory was moved

to a computer program, on which researchers can download the ACT-R code, input information about a specific task, and analyze people's predicted performances.¹²

Consequences

The Adaptive Control of Thought theory, in all its iterations and models, rests on belief in a unitary theory of mind. This theory of mind suggests that we acquire complex skills thanks to an underlying 'human' system that differentiates us from other creatures. Instead of suggesting that we have actually evolved specialized brain faculties, the unitary theory of mind suggests we have an innate neurological component that enables us to successfully acquire more skills as we grow and gain experience. Along these lines, becoming an 'expert' in something is not thanks to a special characteristic unique to an individual, but rather thanks to our brain's ability to effectively maneuver its underlying system. What makes humans so impressive and unique then, is the plasticity (flexibility) of this system, which we can apply to different kinds of skills.¹⁰

ACT therefore provides evidence for skills that humans have acquired that have little to do with evolution. For example, there are many tech-savvy individuals who are great at computer programming. Computer programming would not be a skill anticipated in the evolutionary process, as it does not ensure our survival, and therefore there is no special faculty that enables us to perform that skill. Rather, the ACT theory applies since it suggests that the brain capacity required for computer programming uses the same underlying processes that we use for other skills as well, and we simply have to learn to apply our brain to the new context.

ACT theory also supports the idea that all behavior is a response to stimulus, as put forward by the [behavioral perspective](#). This school of thought suggests that all behavior is learned through conditioning, meaning there must be an underlying cognitive system that is able to learn how to respond to different stimuli (or cognitive units) in our environment.

Controversy

It seems odd that all of our cognitive processes can be reduced to a simple mathematical program or code. ACT theory suggests that all the skills we acquire are thanks to an underlying system of information processing, but it is unclear whether that system is impacted by emotions, cognitive biases, or other internal events.

There are other models of cognitive processes that suggest that our specific brain areas are in fact activated for different purposes, negating the unitary theory of mind. For example, the frontal lobe is thought to be responsible for movement, speech, and reasoning, while the temporal lobe is activated for memory, object recognition and understanding language.¹³ In particular, humans' ability for language acquisition, which differentiates us from other creatures, is widely believed to be due to the existence of specific 'language' areas in the brain. This research suggests that we in fact do have special faculties responsible for different behavior. However, it could be argued that the underlying system proposed by ACT involves all these different parts of the brain.

While the ACT theory explains different higher cognitive processes, it fails to explain why some people are 'experts' and others not, in particular areas. Additionally, while it informs us of the basic system that underlies memory, it does not get into detail about why some cognitive units are remembered better than others. Other theories, like the [levels of processing effect](#), better explain how we use levels to encode information.

Lastly, ACT-R suggests that people always perform rationally. However, according to many behavioral scientists, people actually perform according to [bounded rationality](#). Bounded rationality suggests that our ability to act rationally is impeded by our limited cognitive capacity, time, and available information, all components not factored into ACT theory. In other words, although we ideally act rationally, in practice, we rarely do so.

ACT in Education

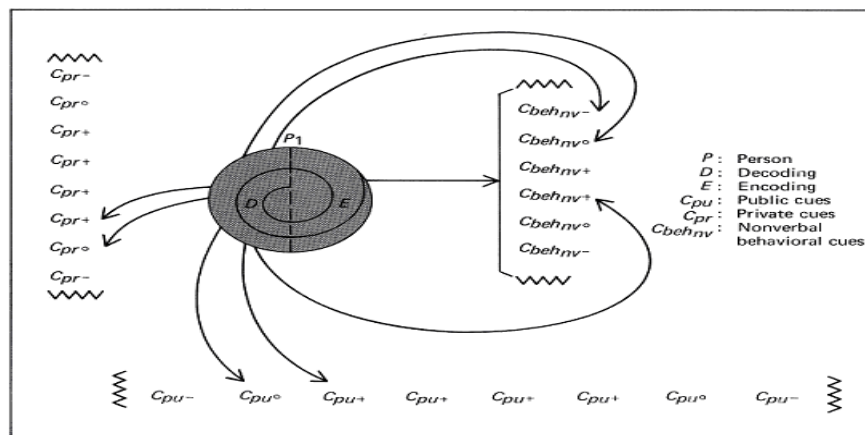
The ACT theory suggests that we process information in cognitive units in our working memory, which distinguishes between declarative and procedural knowledge. The theory also suggests that our working memory can only handle a certain number of cognitive units (5) at a time before becoming overloaded. This information can be important for teachers to know, so that they can break down their material into chunks of knowledge that children can more easily process.²

Since the model also divides knowledge into two distinct kinds, it can inform teachers that the best way to ensure information is encoded and easy to retrieve is making that information both declarative and procedural. Students should be encouraged to combine their knowledge with actions. For example, combining an action or activity with a learned fact means that information will be encoded as both kinds of knowledge and therefore deepen the level of learning.²

Barnlund's Transactional Model of Communication

Dean Barnlund proposed a transactional model of communication in 1970 for [basic interpersonal communication](#) which articulates that sending and receiving of messages happens simultaneously between people which is popularly known as Barnlund's Transactional Model of Communication.

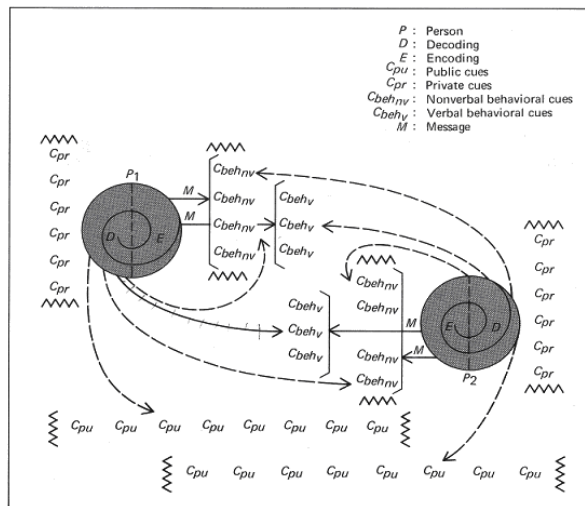
The model has been further adapted and reformed by other theorists as General Transactional Model. The model shifted from the trend of [linear model](#) to dynamic and two way communication model.



Barnlund's Transactional Model is a multi-layered feedback system. This is a continuous process where sender and receiver interchange their places and both are equally important. The message passing takes place with a constant feedback being provided from both parties. A feedback for one is the message for the other.

Components of Barnlund's model

Cues refers to the signs for doing something. As per Barnlund there are: public cues, private cues and behavioral cues. In the model diagram shown above, spiral lines gives graphic representation to the assumptions like public cues and private cues.



- Public cues (Cpu) are physical, environmental or artificial and natural or man-made.
- Private cues (Cpr) are also known as private objects of orientation which include senses of a person. Both these cues can be verbal as well as non-verbal. Another set of cues are behavioral cues.
- Behavioral cues can be verbal (Cbehv) as well as non-verbal (Cbehnv).

The arrows and their directions show that the message is intentionally sent and actively taken where the receiver plays a key role of giving feedback. Arrows also show the process of production of technical encoding, interpretation and decoding.

The jagged lines show that the availability of cues can be unlimited and are denoted as VVVV.

The valence signs, +, 0 and – are also attached to these types of cues which illustrates the amount/degree/strength of attractiveness of the cues in the message.

Speech act refers to particular instance of communication in the model.

Filters are the realities of people engaged in communication. Here the senders' and receivers' personal filters might differ according to cultures, traditions, content of the message, etc.

Noise is the problem that arises in communication flow and disturbs the message flow.

Advantages of Barnlund's Transactional Model of Communication

- The model shows shared field experience of the sender and receiver.
 - Transactional model talks about simultaneous message sending, noise and feedback.
 - Barnlund's model is taken by critics as the most systematic model of communication.
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Disadvantages of Barnlund's Transactional Model of Communication

- Barnlund's model is very complex.
- Both the sender and receiver must understand the codes sent by the other. So they must each possess a similar "code book". (The concept of code book is not mentioned in the model but understood.)

Barnlund's transactional model of communication with Example

The transactional model of communication is a theory that describes communication as a dynamic process where both the sender and receiver of a message are simultaneously encoding and decoding information while engaging in feedback loops. The transactional model was developed by communication theorists Barnlund and Watzlawick in the late 1960s.

In this model, communication is viewed as a two-way process where both the sender and receiver of a message are considered to be active participants in the communication process. Both parties send and receive messages, and the meaning of those messages is shaped by the context in which they are communicated.

The transactional model recognizes that communication is not a linear process but a circular one, with feedback loops between sender and receiver. Communication is not simply about transmitting information from one person to another, but about understanding and negotiating meaning in a shared social context.

The model also recognizes that communication is influenced by various factors, including the environment in which it occurs, the culture and social norms of the communicators, and their backgrounds and experiences.

Overall, the transactional model of communication emphasizes the importance of both verbal and nonverbal communication, the dynamic nature of communication, and the role of context in shaping the meaning of messages.

Barnlund's transactional model

Barnlund's transactional model of communication builds upon the basic transactional model of communication and adds a few more elements to it. This model was developed by Dean Barnlund, an American communication theorist, in the 1970s.

According to the model, communication involves four key elements: the communicators, their communication skills, the context, and the message itself. In this model, communicators are viewed as individuals who bring their unique backgrounds, experiences, and perspectives to the communication process.

Barnlund also emphasizes the importance of communication skills, which include verbal and nonverbal behaviors, listening skills, and the ability to interpret and respond to feedback.

The context in which communication occurs is also an important aspect of Barnlund's model. Context refers to the physical, social, and cultural environment in which communication occurs. The context can affect how a message is perceived and interpreted.

Finally, the message itself is a dynamic process involving encoding, decoding, and feedback. Barnlund's model emphasizes that messages are not simply transmitted from one person to another but are shaped and influenced by the communicators, their skills, and the context in which they communicate.

Overall, Barnlund's transactional model of communication highlights the complexity of communication and the many factors that influence how messages are perceived and interpreted. It emphasizes the importance of communication skills, feedback, and the context in shaping the meaning of messages.

Transactional model of communication example

Suppose two colleagues, Anna and Michael, are working on a project together. Anna emails Michael asking for his feedback on a particular aspect of the project. Michael responds with a detailed critique of Anna's work. In response, Anna feels frustrated and misunderstood.

In this example, Anna and Michael are the communicators. They each bring their unique perspectives and experiences to the communication process. Anna's email message is the initial communication that starts the process, and Michael's response is a feedback message that responds to Anna's initial communication.

Communication skills are also important in this example. Anna's email message may have been unclear or not specific enough, which caused Michael to provide feedback that was not helpful. Similarly, Michael's feedback message may have been too critical, causing Anna to feel frustrated.

The context of the communication is also important. Email communication can be prone to misunderstandings because it lacks the nonverbal cues that are present in face-to-face communication. Additionally, the context of the project itself, such as the deadline or the stakes involved, can affect how Anna and Michael communicate with each other.

Overall, this example illustrates how communication is a complex and dynamic process that involves multiple elements. Barnlund's transactional model of communication highlights the importance of each of these elements and how they interact to shape the meaning of messages.

How to use the Transactional model of communication in our daily lives

The transactional communication model can be used in our daily lives to improve our communication skills and deepen our understanding of how communication works. Here are a few ways to apply the model in your everyday life:

1. Pay attention to nonverbal cues: The transactional model of communication recognizes the importance of nonverbal communication, such as facial expressions, tone of voice, and body language. Paying attention to these cues can help you better understand what someone is trying to communicate.

2. Practice active listening: In the transactional model, communication is a two-way process where both parties send and receive messages. Practicing active listening, paying attention to what the other person is saying, and responding with empathy and understanding can help improve communication and build stronger relationships.
3. Be aware of your own communication skills: The transactional model emphasizes that communication skills are important in shaping the meaning of messages. Reflecting on your own communication skills and working to improve them can help you better convey your thoughts and ideas to others.
4. Consider the context of your communication: The context of communication can affect how messages are perceived and interpreted. Knowing the context can help you better tailor your message to the situation and improve communication.
5. Use feedback loops to clarify understanding: In the transactional model, communication involves feedback loops between sender and receiver. Feedback loops, such as asking for clarification or restating what you heard, can help ensure that both parties are on the same page and understand each other's messages.

By applying the principles of the transactional model of communication to your daily life, you can improve your communication skills and build stronger relationships with others.